

Industry-Oriented Data Cleaning and Preprocessing Problem Statements with Mini Project Extensions

Note: Mini project extensions part is optional. Advanced learners are encouraged to carryout mini project.

1. Transportation Analytics

Title

Vehicle Fleet Data Cleaning and Analysis Using Pandas

Problem Statement

A transportation company maintains fleet records containing:

- VehicleID
- FuelConsumption
- Route

Some records contain missing fuel consumption values and route information.

Perform the following operations:

1. Create a DataFrame containing missing values.
2. Fill missing FuelConsumption values using the mean FuelConsumption.
3. Remove rows where VehicleID or Route is missing.
4. Convert FuelConsumption to integer type.
5. Normalize FuelConsumption using Min-Max Scaling.

Mini Project Extension

Develop a **Fleet Performance Monitoring System** that:

- Identifies fuel-inefficient vehicles.
- Generates route-wise fuel consumption reports.
- Predicts maintenance requirements.
- Monitors fleet utilization.
- Visualizes transportation performance metrics.

2. Manufacturing Analytics

Title

Factory Production Data Preprocessing Using Pandas

Problem Statement

A manufacturing plant maintains production records containing:

- MachineID
- ProductionUnits
- ProductionLine

Perform:

1. Create data with missing values.
2. Fill missing ProductionUnits using the mean value.
3. Remove rows with missing MachineID or ProductionLine.
4. Convert ProductionUnits to integer type.
5. Normalize ProductionUnits using Min-Max Scaling.

Mini Project Extension

Develop a **Smart Manufacturing Dashboard** that:

- Monitors machine productivity.
 - Detects underperforming production lines.
 - Generates production reports.
 - Predicts machine maintenance schedules.
 - Visualizes manufacturing trends.
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3. Sports Analytics

Title

Athlete Performance Data Cleaning Using Pandas

Problem Statement

A sports academy stores athlete records containing:

- AthleteID
- TrainingHours
- Sport

Perform:

1. Create a DataFrame with missing values.
2. Replace missing TrainingHours with the average training hours.
3. Remove records with missing AthleteID or Sport.
4. Convert TrainingHours to integer type.
5. Normalize TrainingHours values.

Mini Project Extension

Develop an **Athlete Performance Analytics System** that:

- Tracks athlete training patterns.
 - Compares sports-wise training intensity.
 - Identifies high-performing athletes.
 - Generates performance reports.
 - Visualizes training statistics.
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4. Renewable Energy Analytics

Title

Wind Turbine Performance Data Preprocessing Using Pandas

Problem Statement

An energy company maintains turbine records containing:

- TurbineID
- PowerOutput
- Location

Perform:

1. Create data with missing values.
2. Replace missing PowerOutput values with average output.
3. Remove incomplete records where TurbineID or Location is missing.

4. Convert PowerOutput into integer format.
5. Normalize PowerOutput values.

Mini Project Extension

Develop a **Wind Energy Monitoring Dashboard** that:

- Tracks power generation efficiency.
 - Identifies low-performing turbines.
 - Generates location-wise energy reports.
 - Predicts maintenance schedules.
 - Visualizes renewable energy trends.
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5. Logistics and Supply Chain Analytics

Title

Warehouse Inventory Data Cleaning Using Pandas

Problem Statement

A logistics company maintains inventory records containing:

- ItemID
- StockQuantity
- Warehouse

Perform:

1. Create a DataFrame with missing values.
2. Fill missing StockQuantity with average stock quantity.
3. Remove rows where ItemID or Warehouse is missing.
4. Convert StockQuantity to integer type.
5. Normalize inventory levels.

Mini Project Extension

Develop an **Inventory Management Dashboard** that:

- Monitors stock availability.
- Identifies low-stock items.
- Generates warehouse-wise inventory reports.
- Supports demand forecasting.

- Visualizes inventory utilization.
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6. Movie Streaming Analytics

Title

Movie Rating Data Preprocessing Using Pandas

Problem Statement

A streaming platform maintains movie records containing:

- MovieID
- Rating
- Genre

Perform:

1. Create a dataset with missing values.
2. Fill missing Rating values using average rating.
3. Remove rows with missing MovieID or Genre.
4. Convert Rating into integer type.
5. Normalize rating values.

Mini Project Extension

Develop a **Movie Recommendation Analytics Dashboard** that:

- Identifies top-rated movies.
 - Generates genre-wise reports.
 - Tracks viewer preferences.
 - Supports recommendation systems.
 - Visualizes movie popularity trends.
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7. Human Resource Analytics

Title

Employee Training Data Cleaning Using Pandas

Problem Statement

An organization maintains employee training records containing:

- EmployeeID
- TrainingHours
- SkillDomain

Perform:

1. Create a DataFrame with missing values.
2. Fill missing TrainingHours using the average hours.
3. Remove records with missing EmployeeID or SkillDomain.
4. Convert TrainingHours to integer type.
5. Normalize TrainingHours values.

Mini Project Extension

Develop a **Corporate Training Analytics Dashboard** that:

- Tracks employee skill development.
 - Generates department-wise training reports.
 - Identifies employees requiring additional training.
 - Visualizes learning progress.
 - Supports workforce planning.
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8. Telecommunication Analytics

Title

Mobile Network Usage Data Preprocessing Using Pandas

Problem Statement

A telecom provider maintains customer usage records containing:

- SubscriberID
- DataUsage
- NetworkRegion

Perform:

1. Create a dataset with missing values.

2. Replace missing DataUsage using average usage.
3. Remove incomplete records.
4. Convert DataUsage into integer type.
5. Normalize DataUsage values.

Mini Project Extension

Develop a **Telecom Usage Analytics Dashboard** that:

- Tracks customer usage behavior.
 - Identifies high-consumption subscribers.
 - Generates region-wise usage reports.
 - Supports network optimization.
 - Visualizes usage trends.
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9. Aviation Analytics

Title

Flight Operations Data Cleaning Using Pandas

Problem Statement

An airline maintains flight records containing:

- FlightID
- DelayMinutes
- Route

Perform:

1. Create a DataFrame with missing values.
2. Fill missing DelayMinutes using average delay.
3. Remove records with missing FlightID or Route.
4. Convert DelayMinutes into integer type.
5. Normalize DelayMinutes values.

Mini Project Extension

Develop a **Flight Operations Dashboard** that:

- Monitors flight delays.
- Generates route-wise performance reports.

- Identifies recurring delay patterns.
 - Supports operational planning.
 - Visualizes flight statistics.
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10. Education Technology (EdTech) Analytics

Title

Online Learning Activity Data Preprocessing Using Pandas

Problem Statement

An online learning platform maintains learner activity records containing:

- LearnerID
- LearningHours
- CourseCategory

Perform:

1. Create data containing missing values.
2. Fill missing LearningHours using mean learning hours.
3. Remove records with missing LearnerID or CourseCategory.
4. Convert LearningHours into integer type.
5. Normalize LearningHours values.

Mini Project Extension

Develop an **Online Learning Analytics Dashboard** that:

- Tracks learner engagement.
 - Generates course-wise participation reports.
 - Identifies at-risk learners.
 - Monitors completion rates.
 - Visualizes learning behavior patterns.
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